



Digital Cameras

The wide acceptance of digital cameras, and the ready availability of software to modify captured images, have prompted newer operating systems such as Windows XP to bundle inbuilt driver support, making the devices virtually plug-and-play.

There are a few features found only in high-end cameras, which will certainly be implemented in entry-level cameras soon. These include the TruePic technology from Olympus, which uses all the pixels even at lower resolutions. The software thus enlarges the image even when it is in raw format on the camera, so as to get a higher interpolated resolution. This reduces the artefacts associated with enlarging photos using software.

Future cameras will also have lenses specifically designed for digital cameras. This is necessary because as the Charge Coupled

Device (CCD) count goes up, the pixel density increases manifold. Even cameras with the best CCDs might not produce a great image if the lens does not let the CCD look through it precisely when zooming in or out.

Today, USB and FireWire are the interfaces of choice for most digital cameras. But devices from laptops to PDAs, and even cell phones, are embracing wireless connectivity to talk with the PC and other mobile devices; the digital camera will surely follow suit.

How A Digital Camera Works

1. Light passes through the lens and is directed to the CCD.
2. The image is focused on the CCD.
3. The ADC (analogue-to-digital converter) converts the image from the CCD into digital signals.
4. The ADC sends data to the DSP



Buying Tips

- ✓ **DIGITAL/OPTICAL ZOOM:** While buying a camera, pay special attention to the optical zoom, especially if you intend to use it outdoors more often. Do not be too impressed by digital zoom.
- ✓ **MEMORY:** Most cameras have 16 MB or less of internal memory. This can store some 30-35 images depending on the subject and resolution, but if you're going to be travelling, you will need external memory cards.
- ✓ **MANUAL SETTINGS:** Buy a camera that has manual exposure and shutter-speed settings. This will allow you to improve indoor shots, where lighting is usually poor.
- ✓ **BUNDLED CABLES:** Check for the necessary USB and video cables. Opt for a

camera that does not need a proprietary USB cable—these are generally expensive and difficult to procure.

- ✓ **AESTHETICS:** Buy a camera that offers good grip and has proper placement of buttons. Slim cameras look good, but they are poor as far as ergonomics and functionality goes.
- ✓ **RECHARGEABLE BATTERIES:** Digital cameras, especially those with an inbuilt LCD screen and optical zoom, require a lot of power. Buy cameras with rechargeable batteries.
- ✓ **ENVIRONMENT PROTECTION:** Some cameras offer splash-resistant packaging, which minimises risk of damage by water or moisture.

Ready Reckoner

	To be able to take photos to compile personal albums or publish to the Web
You Need	
Look for	Between 640 x 480 and 1,024 x 768 resolution and 2 MB memory; perhaps one that doubles up as a Web cam
	High quality images with plenty of configurability
You Need	
Look for	1,280 x 1,024 resolution with at least 2X optical zoom and 8 MB of storage
	Very high quality photographs and expandability
You Need	
Look for	2,048 x 1,536 or higher resolution with special lens options, and a minimum of 16 MB of memory

SONY

(Digital Signal Processor) which creates the image and compresses it.

5. The final image is stored either on the inbuilt flash memory or on a memory stick.

Do Remember

CMOS-based sensors are less expensive to manufacture than CCD-based sensors. But they cannot capture motion well. Many industry analysts believe that eventually, all entry-level digital cameras will be CMOS-based, and that only mid-range and high-end units will use CCDs.

Today you can buy photo printers that can print directly from a digital camera.

Facts

CCD technology was initially developed in the 1960s as a memory device. It was only in 1974 that Fairchild Electronics developed an imaging device using CCDs, with a format of 100 x 100 pixels. Now some high-end cameras have a 3-CCD setup to separately capture red, green and blue light for optimum results.